

DevOps Principles for Network Engineers

DAY 1 - MODULE 1

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What is DevOps?

03

Traditional vs Modern Operations

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GitOps & Infrastructure as Code

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Why Network Teams Need DevOps

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Network Automation Journey

Learning Objectives

By the end of this session, participants will be able to:



Define DevOps

In the context of networking and network operations.



Identify Challenges

Recognize pain points in traditional network operations.



Understand Automation

Explain the importance and value of automation at scale.



Explain IaC

Describe Infrastructure as Code and Git-based workflows.



Describe NetDevOps

Articulate the benefits of applying DevOps to networking.

The Evolution of Network Operations

"Can we manage 500 devices manually in the cloud era?"

Traditional Networking Era

- Device-by-device management
- CLI-based configuration
- Manual troubleshooting
- Spreadsheet-based tracking
- Change requests through tickets

Modern Networking Era

- Automated deployments
- API-driven devices
- Version-controlled configurations
- Continuous validation
- Self-service infrastructure

What is DevOps?

DevOps is a **culture, practice, and set of tools** that combines Development, Operations, Automation, and Collaboration — delivering changes **faster, safer, and more reliably**.

Define Desired State

Describe what the network should look like in code.

Store in Git

Version-control all configurations and templates.

Automate Deployment

Push changes consistently across all devices.

Validate Automatically

Confirm the network state after every change.



Why Traditional Network Teams Struggle

Common Challenges

- Hundreds of devices across multiple vendors
- Manual CLI configuration for every change
- Complex, slow change processes
- Lack of standardization across the environment

Scaling Problem: NTP Server Change

5 devices

Easy

50 devices

Difficult

500 devices

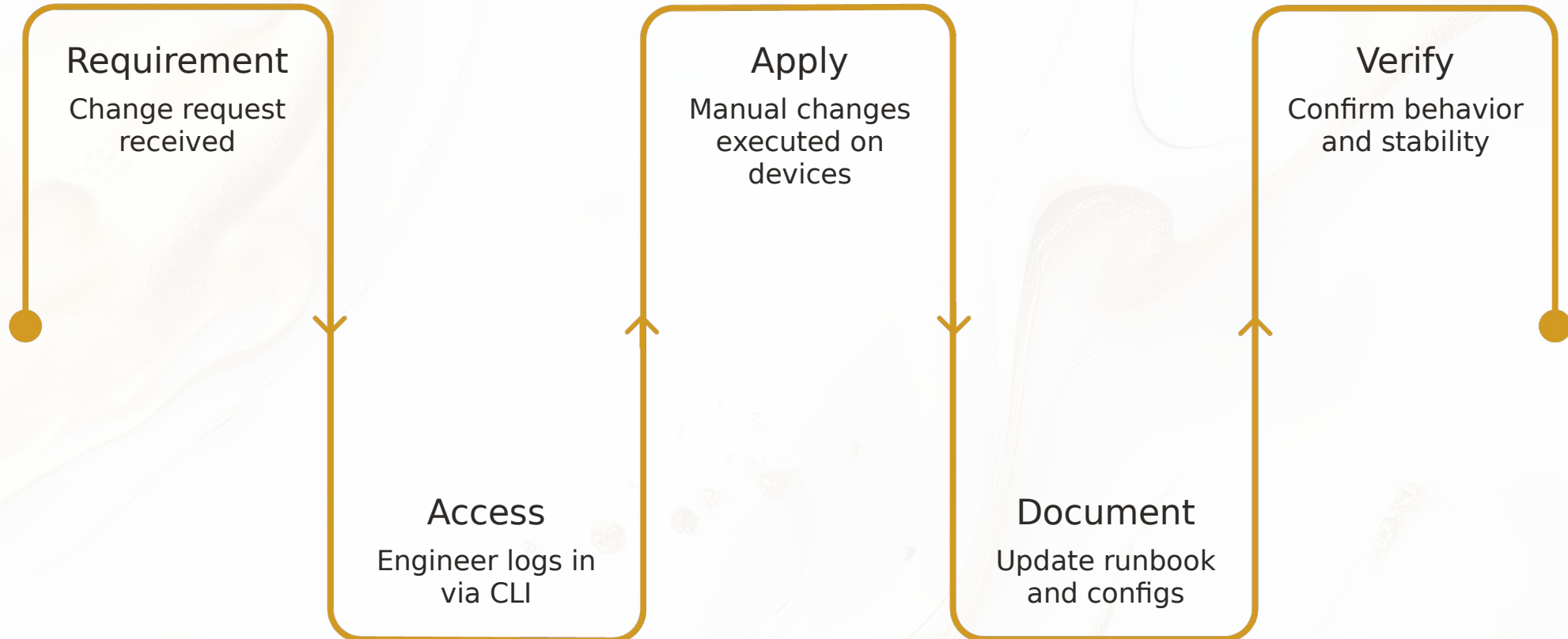
Risky



Manual effort does not scale.

Traditional Network Change Workflow

Every change follows a slow, repetitive, and error-prone manual process:



An engineer logs into Router 1, Router 2, Router 3, Router 4, and Router 5 — repeating the same commands manually on each device. This approach is time-consuming and introduces significant risk at every step.



Problems with Traditional Operations

Human Errors

Wrong VLAN IDs, incorrect IP addresses, ~~missing commands~~, or misconfigured interfaces are common and costly.

No Rollback

Restoring a previous state is difficult. Backups may be outdated or incomplete.

No Audit Trail

No reliable record of who changed what, or when the change was made.

Slow Deployments

Simple changes can take ~~hours or even days~~ to propagate across the environment.

Real Incident: The Cost of One Typo

The Scenario

An engineer updates an ACL across 50 routers. One router receives:

```
deny ip any any
```

instead of:

```
permit ip any any
```

The Impact

- Application outage
- User complaints
- Emergency troubleshooting

Root Cause

Manual configuration error — a single mistyped command on one of 50 devices.

The Lesson

Humans make mistakes. Automation reduces them. A pipeline would have caught this before deployment.

The DevOps Approach to Network Changes

Replacing the manual workflow with an automated pipeline transforms how changes are delivered:



Define Requirement

Git Commit

Peer Review

Trigger Pipeline

Every change becomes **repeatable, traceable, and automated** — eliminating the risk of manual error and providing a full audit trail from requirement to validation.

Understanding Git in Networking

What Git Stores

- Router configurations
- Switch templates
- Firewall policies
- Automation scripts

Key Benefits

- Full version history of every change
- Rollback to any previous state
- Peer review before deployment
- Complete change tracking

 Every configuration version is recorded: **router-config-v1** → **router-config-v2** → **router-config-v3**. Nothing is ever lost.



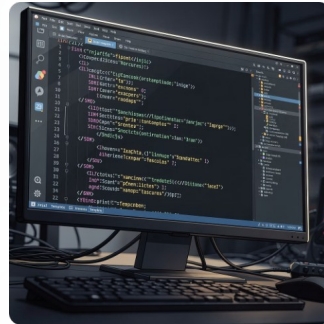
Infrastructure as Code (IaC)

IaC means managing infrastructure through **code rather than manual changes** — replacing one-off CLI sessions with structured, reusable definitions.



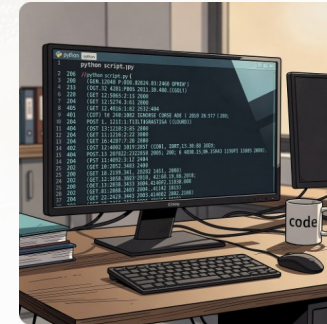
YAML & JSON

Human-readable data formats for defining device configurations and desired states.



Templates

Reusable configuration blueprints applied consistently across all devices.

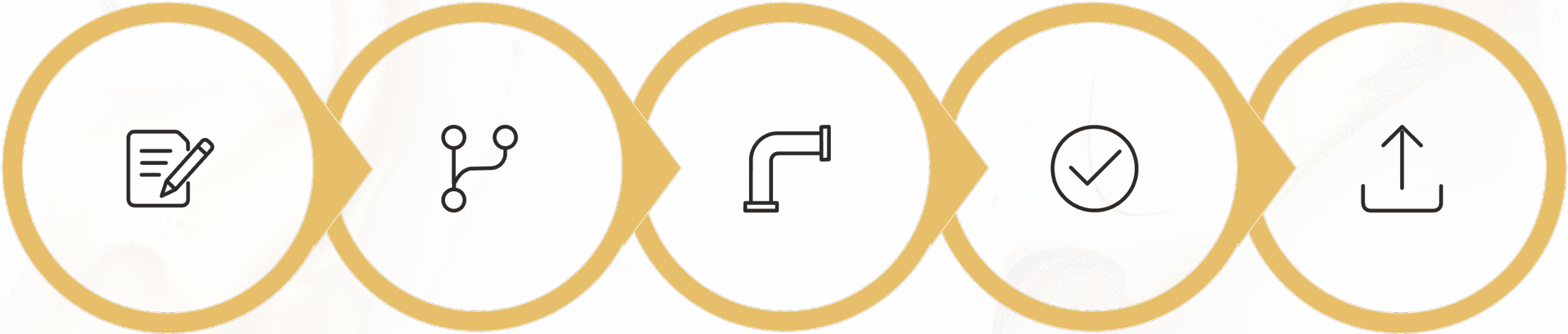


Python & Ansible

Automation scripts and playbooks that execute configurations at scale.

Network Automation Pipeline

A CI/CD pipeline connects the engineer's change to the network automatically:



Update
Template

Git Push

Run CI/CD

Validation

Deploy

GitHub / GitLab

Jenkins

Ansible

Python

Benefits of DevOps for Networking



Repeatability

Same configuration applied every time, across every device.



Auditability

Complete, timestamped change history for every modification.



Faster Deployments

Changes that took hours now complete in minutes.



Compliance

Standardized configurations enforce policy automatically.



Reduced Human Error

Automated execution eliminates manual mistakes.

Traditional Networking vs. NetDevOps

Traditional Networking

Manual CLI

Device-by-device

Limited visibility

Slow deployments

Human dependent

Difficult rollback

NetDevOps

Automated

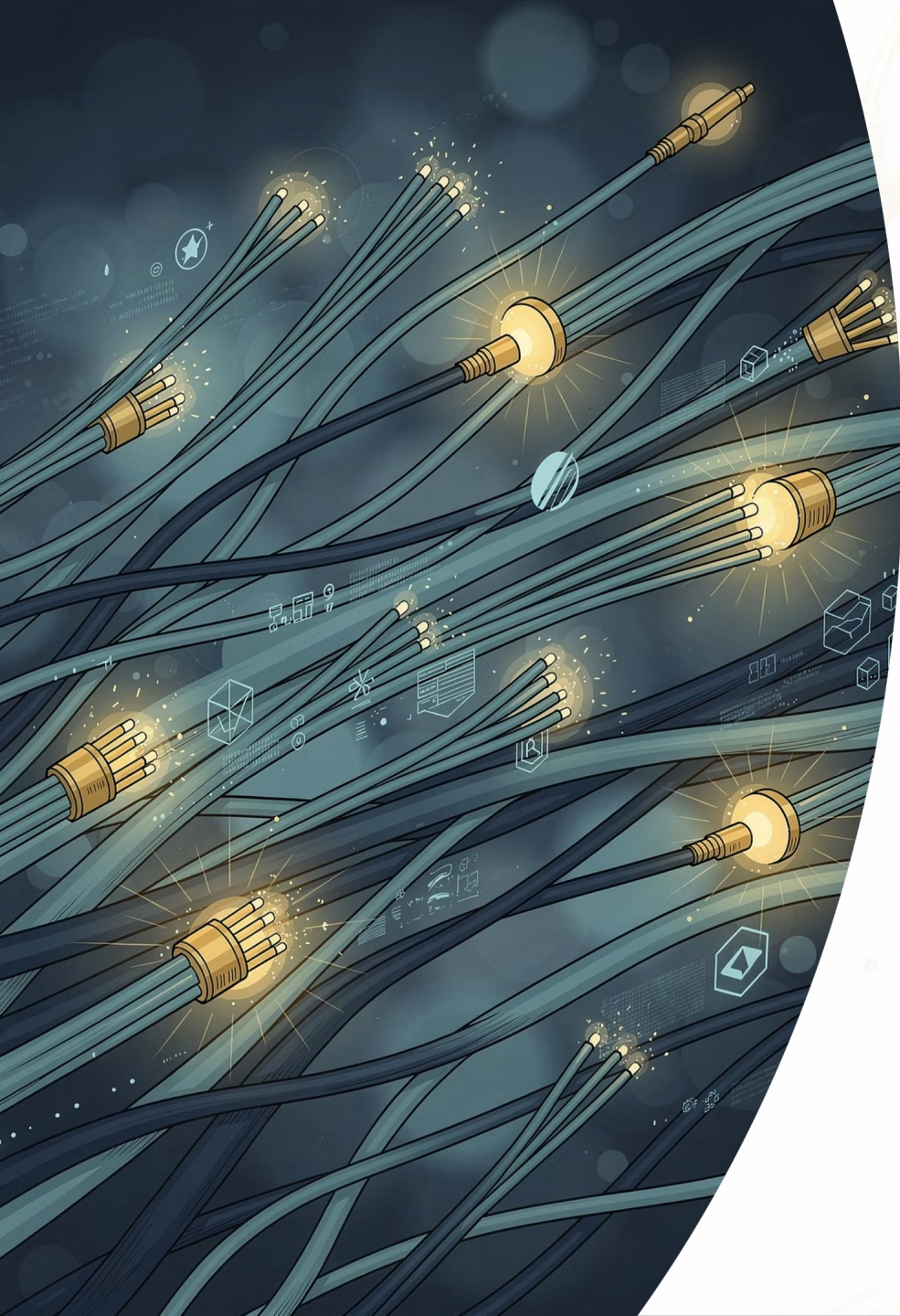
Template-driven

Full audit trail

Fast deployments

Process driven

Easy rollback



CHAPTER BREAK

The Automation Advantage

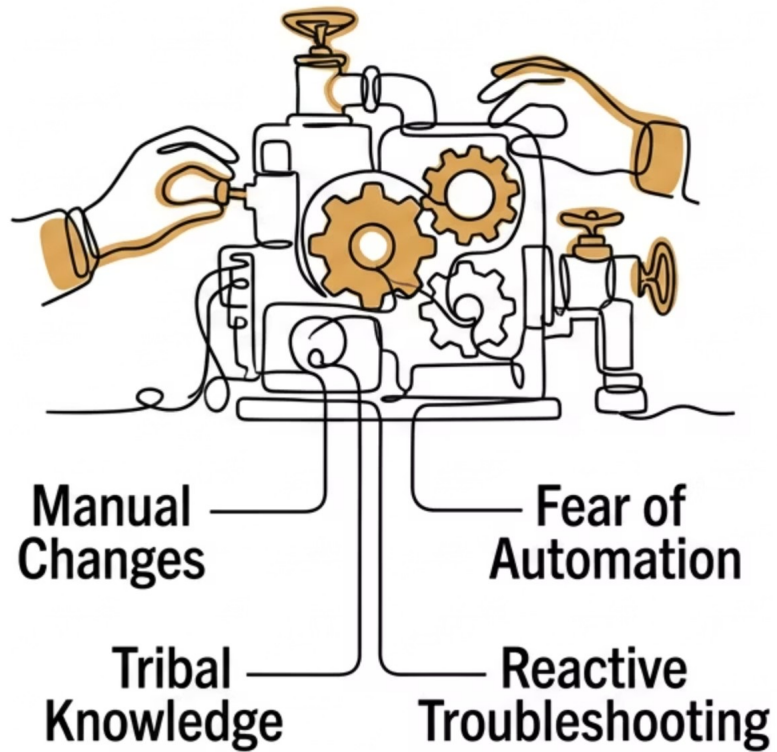
From manual CLI sessions to software-driven networks — the shift is already underway.

Key Takeaways

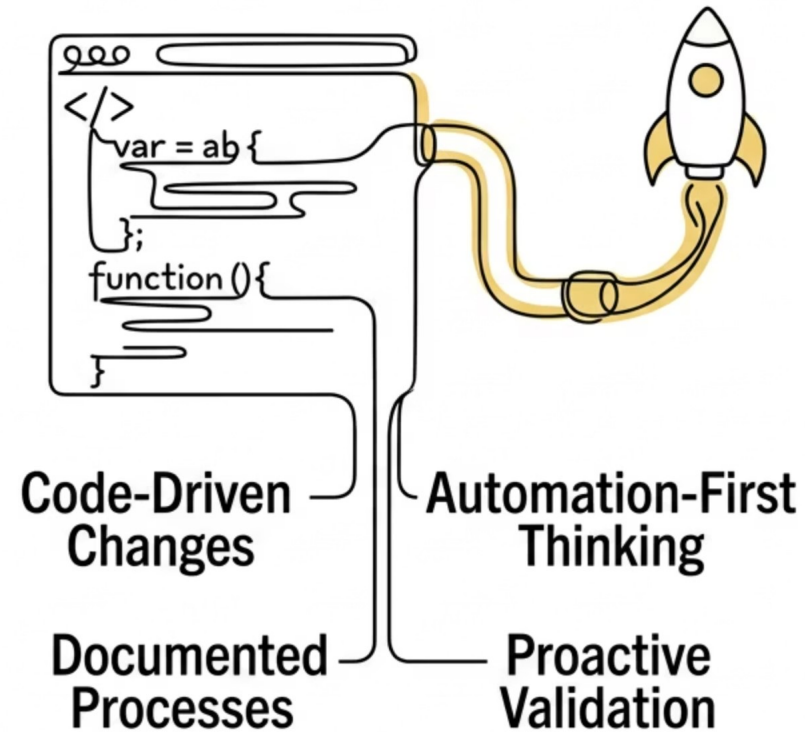
- 1 Networks are becoming software-driven**
The industry is moving toward programmable, API-driven infrastructure.
- 2 Manual operations do not scale**
Managing hundreds of devices by hand is risky and unsustainable.
- 3 Git provides version control**
Every configuration change is tracked, reviewable, and reversible.
- 4 Automation improves consistency**
Pipelines apply the same change correctly every time.
- 5 DevOps enables faster, safer changes**
Peer review and validation catch errors before they reach production.
- 6 NetDevOps is the future**
Combining networking expertise with DevOps practices is the path forward.

The NetDevOps Mindset Shift

TRADITIONAL MINDSET

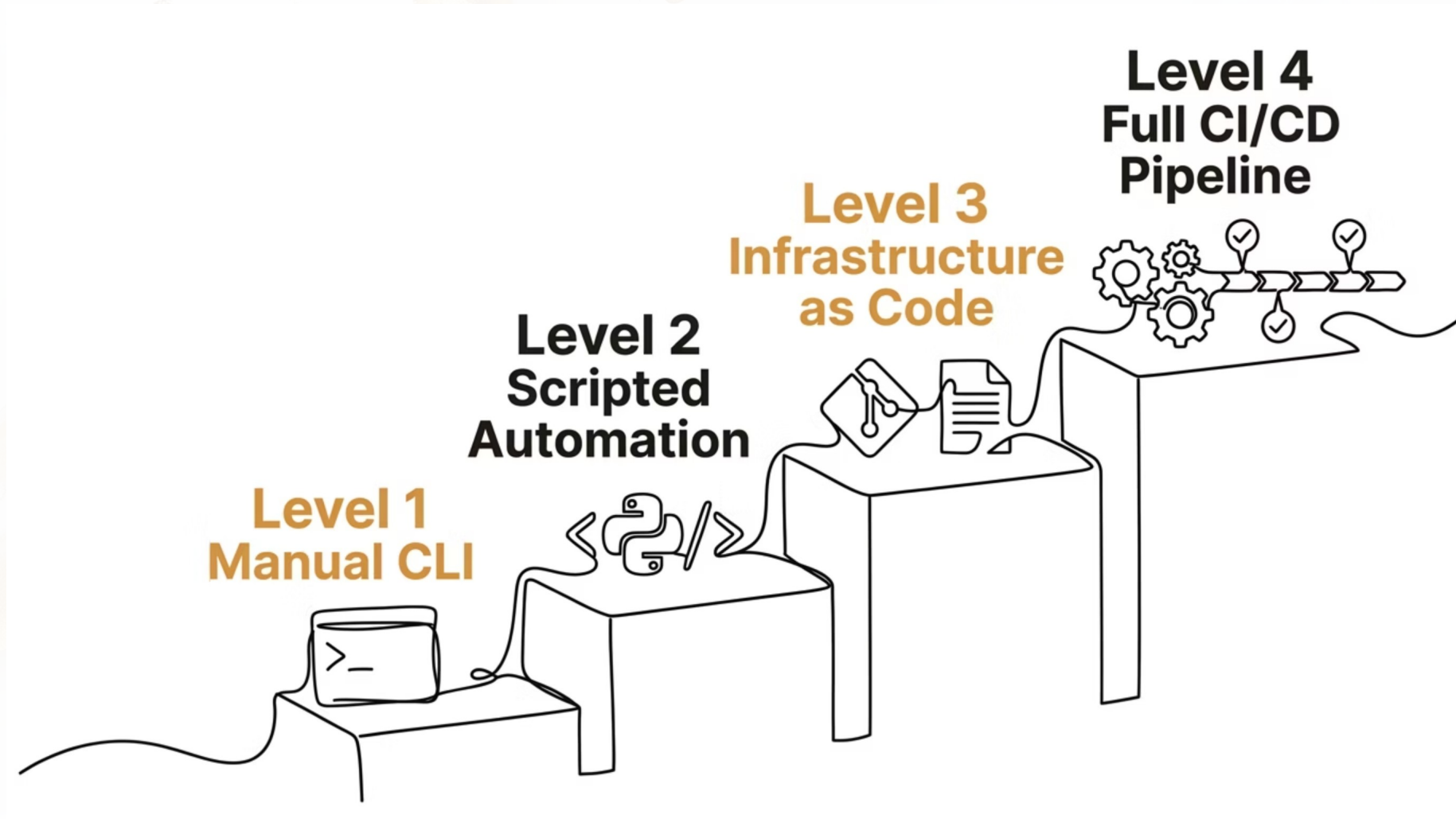


NETDEVOPS MINDSET



Adopting NetDevOps is not just about tools — it requires a fundamental shift in how network engineers think about change, risk, and repeatability.

The Network Automation Journey



Most organizations begin at Level 1. The goal of this course is to equip you with the skills to progress through each level systematically.

Tools of the NetDevOps Engineer



Git & GitHub / GitLab

Version control and collaboration platform for all configurations and automation code.



Ansible

Agentless automation tool for deploying configurations across network devices at scale.



Python

The primary scripting language for network automation, API interaction, and data parsing.



Jenkins / CI/CD

Pipeline orchestration tools that automate testing, validation, and deployment workflows.

What's Next: Module 2

 COMING UP

Git Fundamentals for Network Engineers

In the next module, we dive hands-on into Git — the foundation of every modern NetDevOps workflow.

Version Control

Track every configuration change with full history.

Branching

Safely develop and test changes in isolation.

Change Management

Use pull requests and peer review for network changes.

